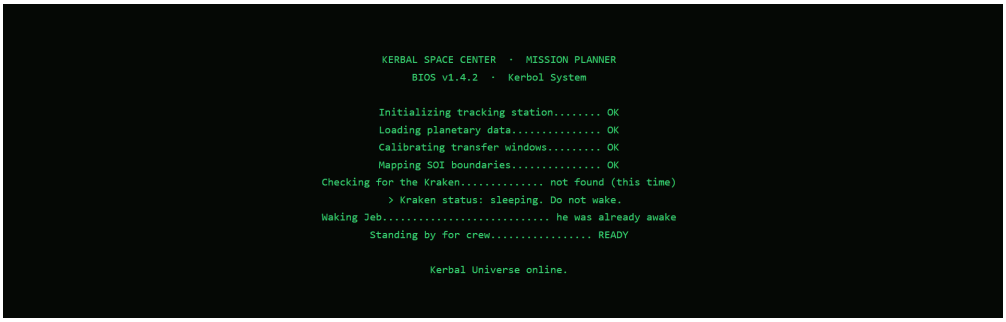
DELTA V PLANNER

Instruction Guide



Open the HTML file in any browser. No install required.

About this tool

The Delta V Planner is a single-file mission planner for Kerbal Space Program. You open the HTML file in any browser — no install. It budgets fuel, power, and antenna before you build the vessel, and it plans resonant relay constellations.

It is a planning tool, not a flight computer. If you fly the ejection, phase angle, and capture close to the numbers (by hand or with MechJeb), you will be in the right ballpark. Small differences between the planner and a flown mission are normal. The goal is to not fall short on fuel, power, or antenna, while not carrying a huge surplus.

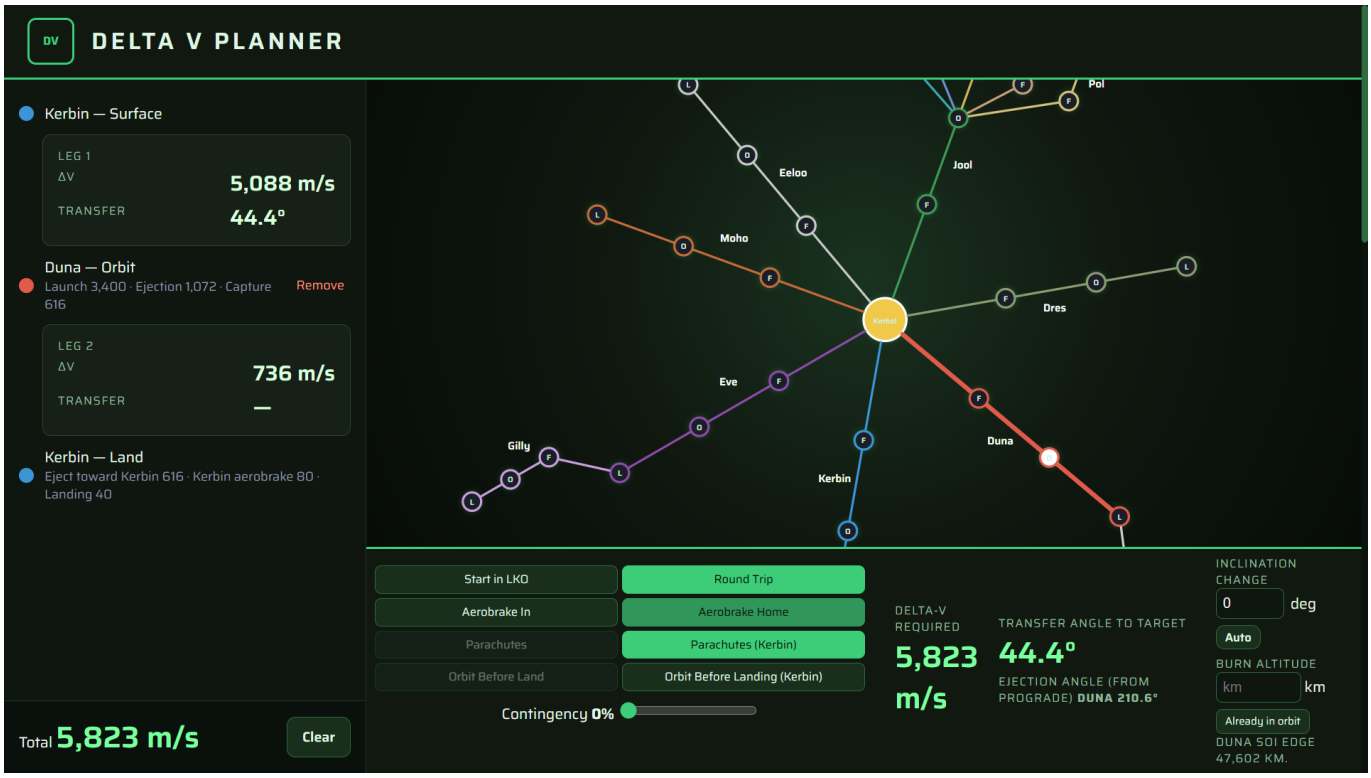
What is accurate

- Hohmann transfers, ejection burns, and capture burns use real KSP patched-conic math and stock body constants (μ, radius, SMA, SOI).
- Phase angles and ejection angles are the standard ideal-Hohmann numbers.
- Plane-change cost on SOI entry uses the correct $2 \cdot v \cdot \sin(\Delta i/2)$ formula at the altitude you set.

What is approximate

- Aerobrake is a flat 80 m/s. Real cost depends on entry speed, periapsis, and how many passes you take.
- Parachute landing is a flat 40 m/s (340 without). Real chute staging varies.
- Fly-by returns use a small fixed correction. A real fly-by depends on your exit path.
- No gravity assists, no launch-window drift, and mid-course corrections are not included.

1. Pick the destination



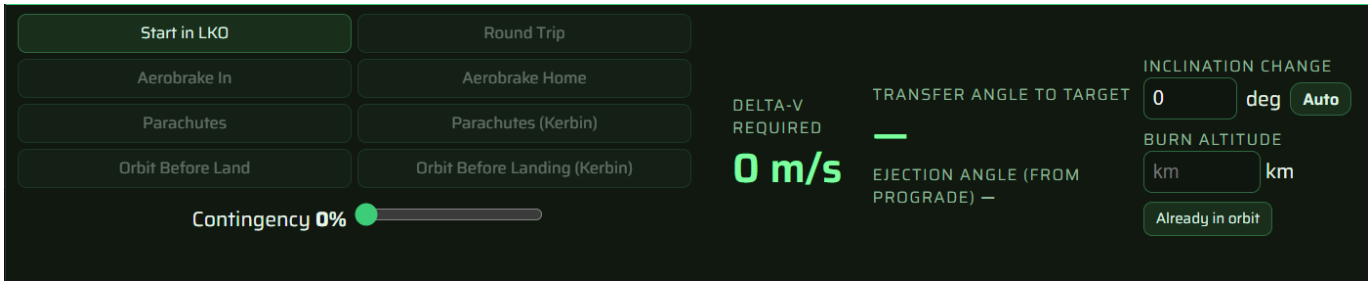
Click a body on the map. The journey builds on the left.

Click the body on the map. For a moon, click the parent first (Duna, then Ike). The left column adds the stop with its delta-v breakdown. Click the stop's marker to cycle the mode:

- **Orbit** — arrive and circularize.
- **Land** — go to the surface. Airless bodies can land direct or orbit first.
- **Fly-by** — pass through the SOI. No capture burn.

Use Remove on a stop, or Clear at the bottom, to start over.

2. Set the mission options



The toggle bar. Hover any button for a tip, even if it is locked.

- **Start in LKO** — already in low Kerbin orbit. Removes the launch burn.
- **Round Trip** — adds the return to Kerbin. Available only after a destination is selected.
- **Aerobrake In** — capture with the destination atmosphere. Only if a body on the route has air.
- **Parachutes** — cut landing delta-v at the destination. Only for a landing on a body with air.
- **Parachutes (Kerbin)** — cut the Kerbin landing burn. Defaults on for a round trip.
- **Orbit Before Land** — circularize at the destination before landing. Off = direct surface entry.
- **Orbit Before Landing (Kerbin)** — circularize at Kerbin on the return before landing.
- **Contingency** — extra percent on the non-launch burns.

3. Choose how you return to Kerbin

A return is always a landing. How you arrive changes the cost:

- **Direct to the surface** — leave Orbit Before Landing (Kerbin) off. Aerobrake Home is forced on. The atmosphere does the capture. Cheapest.
- **Orbit first, with aerobrake** — turn on Orbit Before Landing (Kerbin) and leave Aerobrake Home on. Aerobrake into orbit, then a small burn to stop reentry. Use this to pick a landing site.
- **Orbit first, full burn** — Orbit Before Landing (Kerbin) on, Aerobrake Home off. Full propulsive capture. Use this if you do not want to touch the atmosphere.

Aerobrake Home only becomes toggleable when Orbit Before Landing (Kerbin) is on.

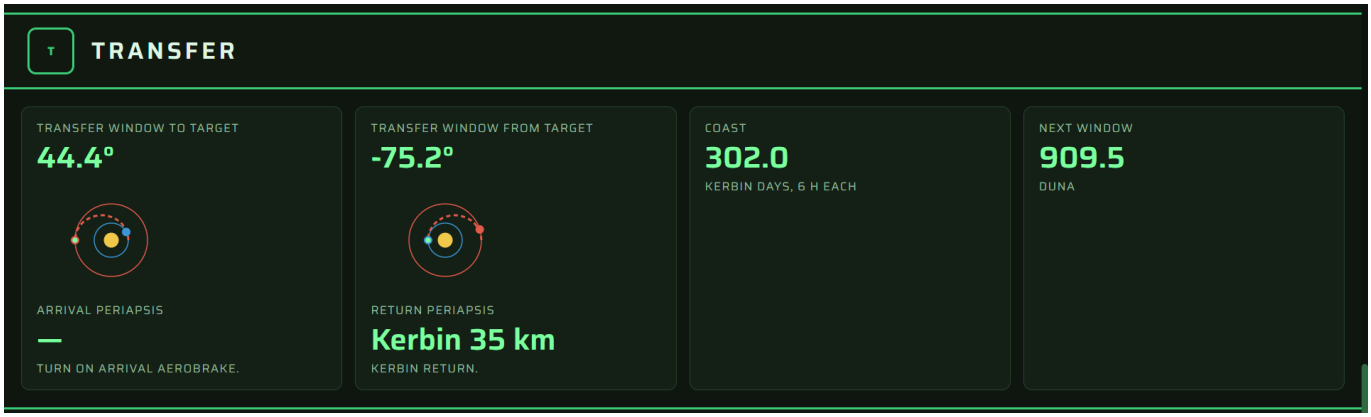
4. Inclination change

Use this when you need a specific orbit at the destination — for example 90° polar for scanning — that differs from your arrival.

- Type the degrees, or press **Auto** to fill the body's inclination.
- Set the burn altitude. Higher is cheaper. The SOI edge is shown under the field.
- **Default** — plane change on SOI entry, before capture. You already know the inclination. Burn far out on the way in.
- **Already in orbit** — the rare case. Raise from low orbit, change plane at apoapsis, drop back. Costs more.

Enter altitudes as 32600 or 32,600. Both work.

5. Read the transfer



Phase angles, coast time, and days until the next window.

- **Transfer angle to target** — the phase angle to wait for.
- **Ejection angle (from prograde)** — where to point the burn relative to Kerbin prograde.
- **Coast** — travel time in Kerbin days (6 hours each).
- **Next window** — days until the next chance.
- The diagrams show a prograde Hohmann arc. The vessel leaves the departure planet and meets the destination. The inner planet moves faster than the outer.

Delta-v required in the bottom bar is the budget for the whole mission.

6. Heat shields and stages



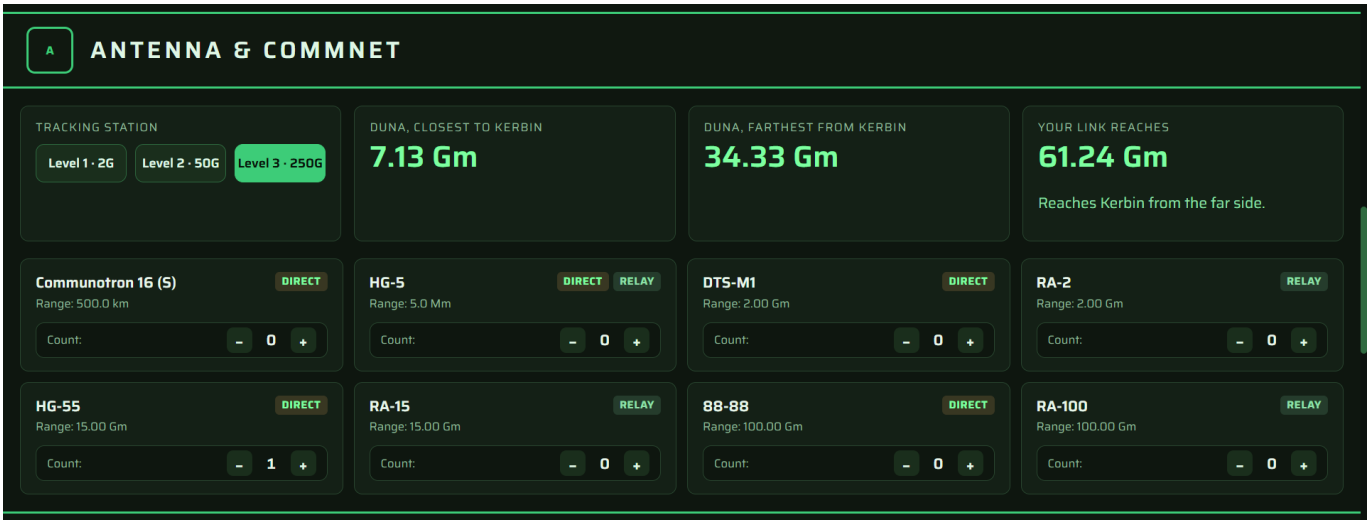
Stages breaks the mission into Launch, Transfer, Lander, and Return, and notes where a heat shield is needed.

The Stages section splits the budget and flags thermal protection.

- **Aerobrake at the destination** — the arriving ship needs a heat shield.
- **Aerobrake Home on** — the return craft needs a heat shield for Kerbin entry.
- Direct surface returns and aero-to-orbit returns both hit the atmosphere. Only a full propulsive capture (Aerobrake Home off) skips the shield.
- Size the shield for the entry speed. The planner warns if Kerbin entry is fast enough that one pass may not be enough.

Direct surface returns and aero-to-orbit returns both hit the atmosphere. Only a full propulsive capture skips the shield.

7. Antenna & CommNet



Tracking Station level, closest and farthest range, and whether your antenna reaches.

Pick the Tracking Station level and set antenna counts. The section shows whether you can reach the destination at closest and farthest approach. Use it so the vessel is not deaf at the target. Relay antennas extend the network; direct antennas talk to Kerbin.

8. Science transmission power

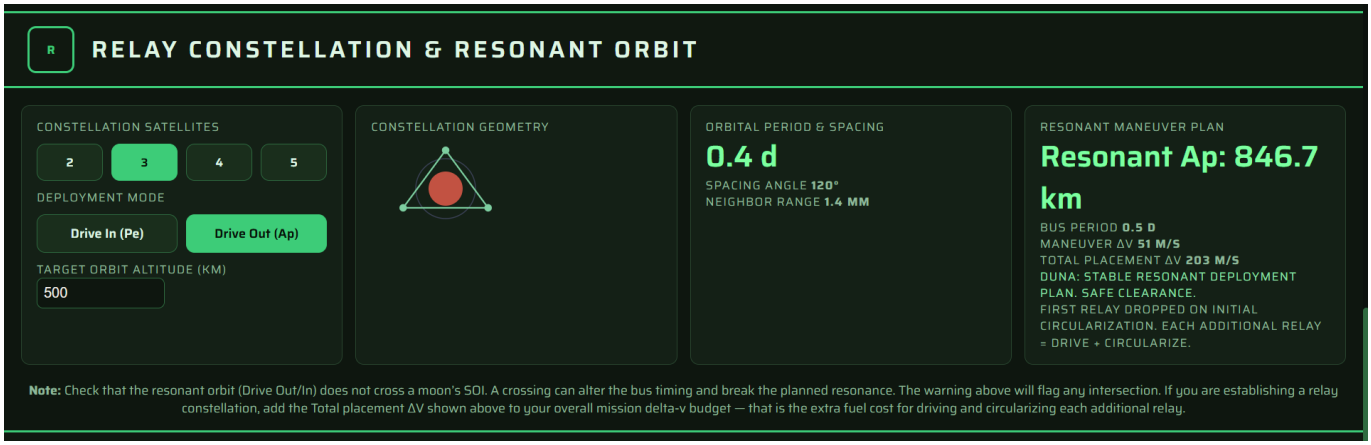


EC cost to transmit each experiment with the selected antenna.

Shows the EC cost to transmit one experiment (or an M700 scan) with the selected antenna. Only experiments that can be run at the destination are listed. These figures are the peak transmission cost only — they do not include the probe core, reaction wheels, or lights. Size batteries for the peak plus your normal load, and account for every experiment you plan to run.

9. Relay constellation

Use this when you want relays around a body. Select the body on the map first.



A 3-satellite constellation: geometry, spacing, and the resonant bus plan.

- Choose **2, 3, 4, or 5** satellites. Default is 3.
- Choose **Drive In (Pe)** or **Drive Out (Ap)**.
- Leave altitude blank for the auto value, or type your own.
- The first relay is dropped on the initial circularization. Each additional relay costs a drive plus a circularize.
- Add the Total placement delta-v to your mission budget.

Watch for moon SOI crossings



Warning: the resonant orbit crosses Ike's SOI. Raise or lower the altitude to clear it.

If the resonant orbit crosses a moon's SOI, the timing can break. The warning names the moon and the altitude band to avoid.

Polar pair (2 satellites)



Two satellites for polar coverage. They cannot see each other.

The 2-satellite case is for polar coverage. The body blocks the direct link between them. Add an equatorial relay if you need them connected. The resonant bus math still runs.

10. Worked example — Duna orbit and return

- Click Duna. Set the mode to Orbit.
- Turn on Round Trip. Parachutes (Kerbin) comes on by default.
- Leave Orbit Before Landing (Kerbin) off for a direct aerobrake home.
- Read Delta-v required, the transfer angle, and the ejection angle.
- Check Antenna so you can reach Duna at farthest approach.
- Check Science Transmission Power and account for every experiment you plan to run.
- Note the heat shield for the Kerbin return.
- Build to that budget and fly the window.

Delta V Planner — plan the budget, then fly it.